

504MAN RESOURCE MANAGEMENT

Decision-Making Tools

Week 4 – Session 16

Register



Intended Module Learning Outcomes (LO's)

By the end of the module students should be able to:

LO1: Appraise the development of the human resource function.

LO2: Critically examine the traditional and contemporary approaches of recruitment and selection.

LO3: Analyse the procedures and practices of performance management and remedial activities.

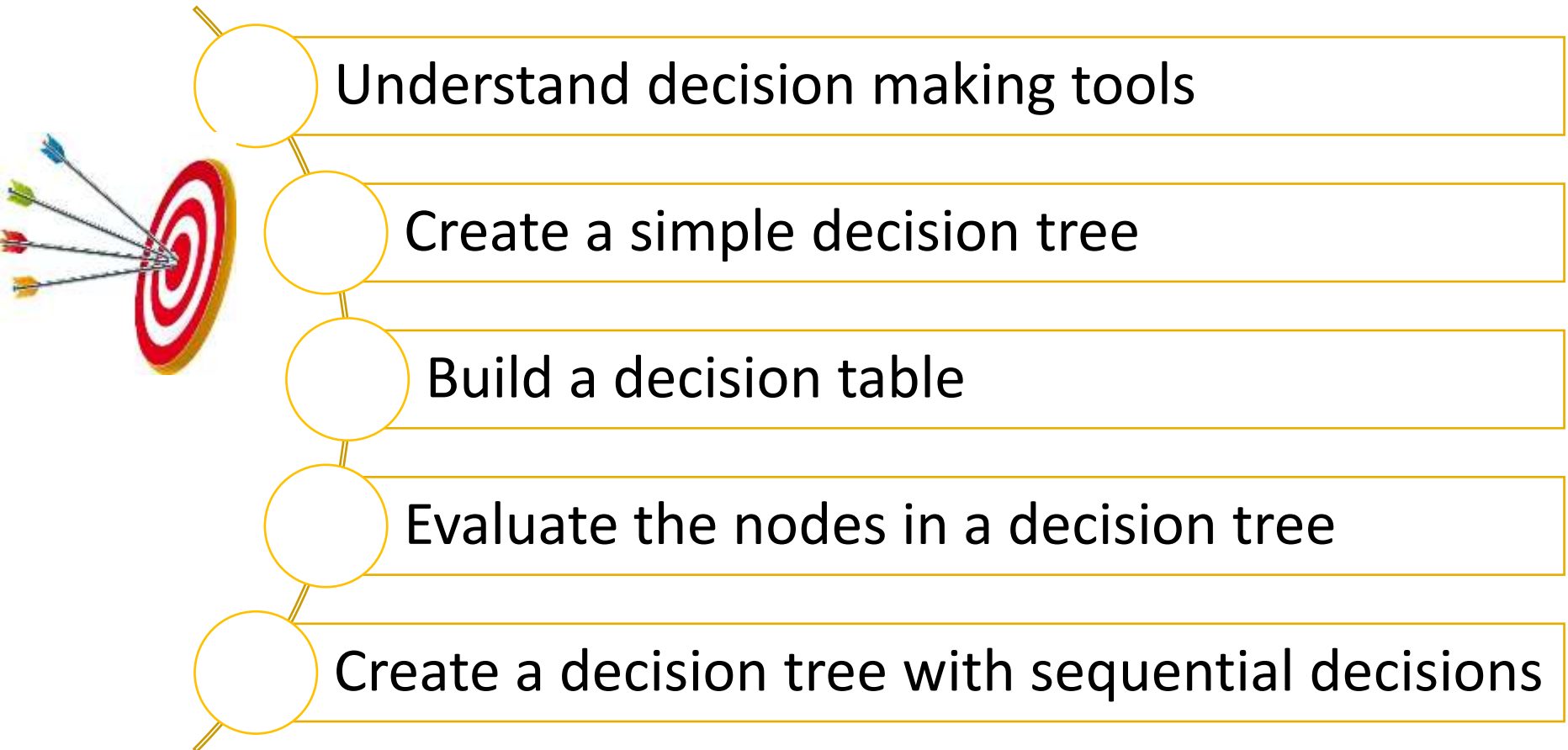
LO4: Investigate and plan the resources required to achieve an organisation's aims and objectives.

LO5: Evaluate and appraise the use of resources against organisational plans.

LO6: Formulate plans to ensure that resources are maximised and remedial actions taken where appropriate.

Session Outcomes

At the end of this session learners should be able to:



Think, Pair & Share



Activity

15 minutes

- What makes the difference between a good decision and a bad decision?

Try to give
real-life
examples

Share your
answers
with the
class



Watch & Think-Pair-Share



Activity

30 minutes

<https://www.youtube.com/watch?v=542qgGgL1s4>

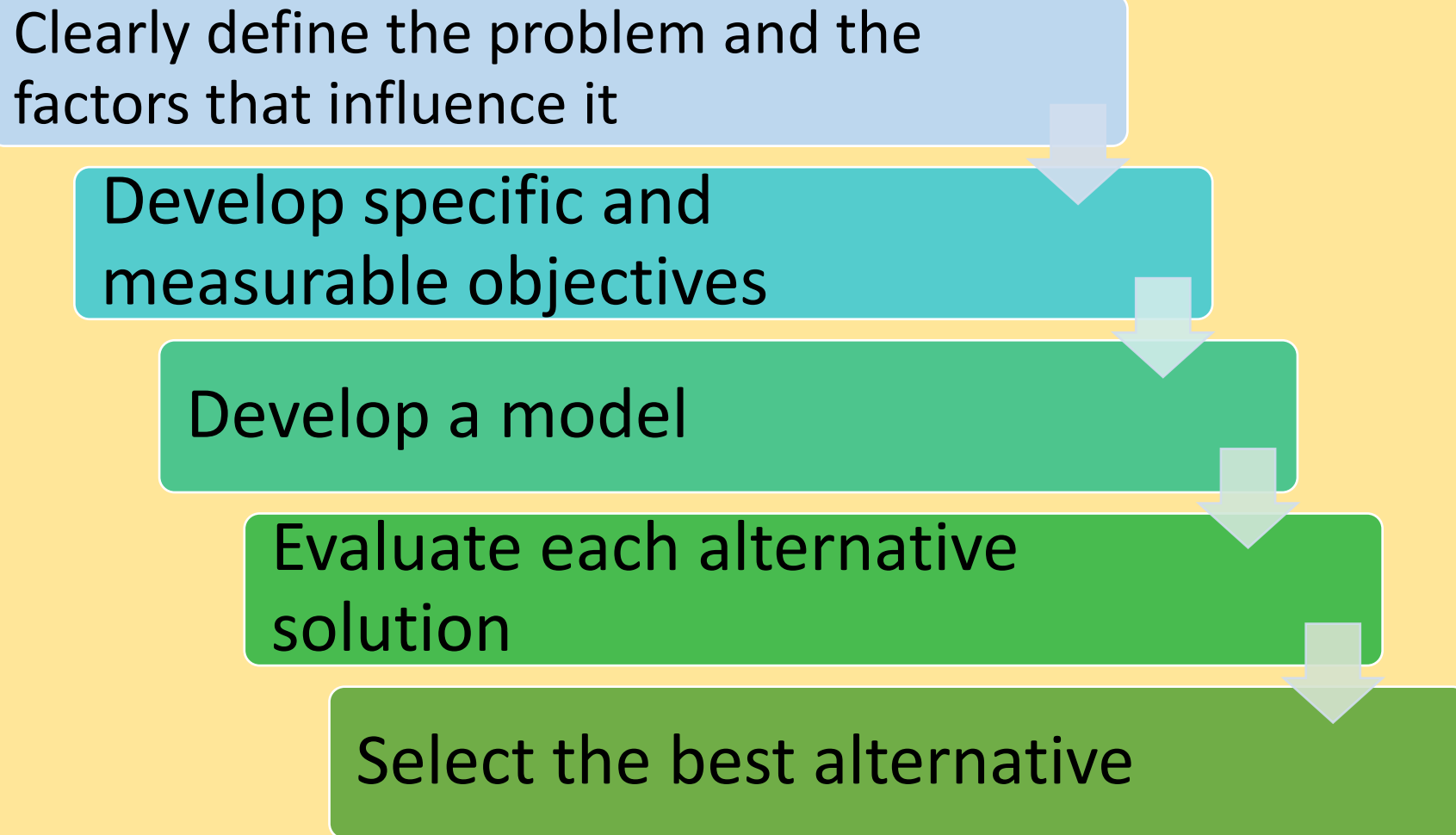


THE POWER OF DECISION-MAKING | BENEDIKT AHLFELD | TEDxGraz

The Decision Process in Operations

A “good” decision—one that uses analytic decision making—is based on logic and considers all available data and possible alternatives. It also follows these six steps:

Clearly define the problem and the factors that influence it



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graph TD; A[Clearly define the problem and the factors that influence it] --> B[Develop specific and measurable objectives]; B --> C[Develop a model]; C --> D[Evaluate each alternative solution]; D --> E[Select the best alternative];
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Develop specific and measurable objectives

Develop a model

Evaluate each alternative solution

Select the best alternative

7 STEPS TO EFFECTIVE DECISION MAKING

Decision making is the process of making choices by identifying a decision, gathering information, and assessing alternative resolutions.

Using a step-by-step decision-making process can help you make more deliberate, thoughtful decisions by organizing relevant information and defining alternatives. This approach increases the chances that you will choose the most satisfying alternative possible.

①

**IDENTIFY
THE DECISION**

②

**GATHER
INFORMATION**

③

**IDENTIFY
ALTERNATIVES**

④

**WEIGH THE
EVIDENCE**

⑤

**CHOOSE
AMONG
ALTERNATIVES**

⑥

TAKE ACTION

⑦

**REVIEW YOUR
DECISION**

Fundamentals of Decision Making

1. Terms:

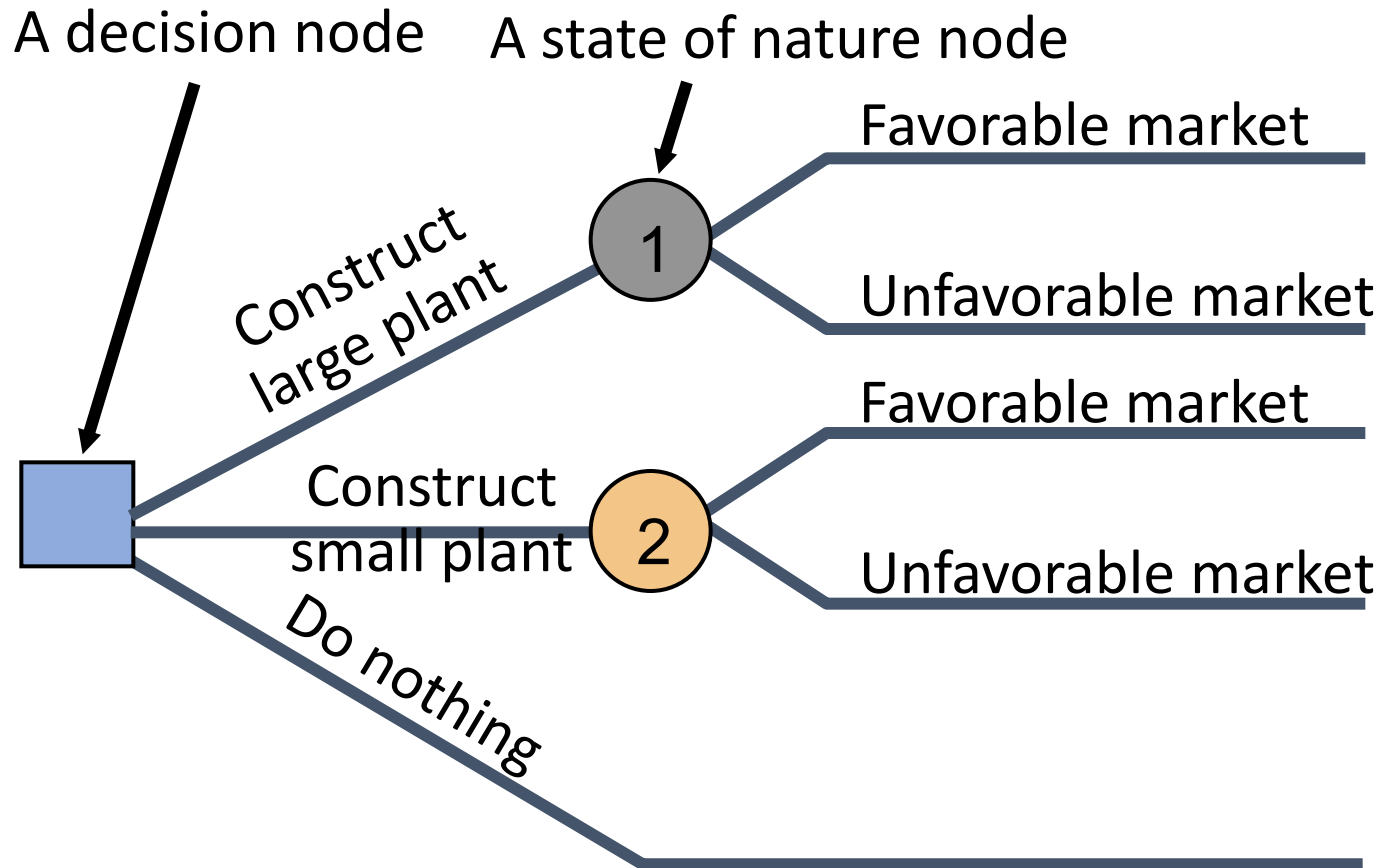
- a. Alternative* – a course of action or strategy that may be chosen by the decision maker
- b. State of nature* – an occurrence or a situation over which the decision maker has little or no control

Fundamentals of Decision Making

2. Symbols used in a decision tree:

- a. □ – Decision node from which one of several alternatives may be selected
- b. ○ – A state-of-nature node out of which one state of nature will occur

Decision Tree Example



Decision-Making Environments

Decision making under uncertainty

- Complete uncertainty as to which state of nature may occur

Decision making under risk

- Several states of nature may occur
- Each has a probability of occurring

Decision making under certainty

- State of nature is known

Uncertainty

1. Maximax
2. Maximin
3. Equally likely

1. Maximax

- Find the alternative that *maximizes* the *maximum* outcome for every alternative
- Pick the outcome with the maximum number
- *Highest* possible *gain*
- This is viewed as an optimistic decision criteria

Uncertainty

2. Maximin

- Find the alternative that *maximizes* the *minimum* outcome for every alternative
- Pick the outcome with the minimum number
- *Least possible loss*
- This is viewed as a pessimistic decision criteria

Uncertainty

3. Equally likely

- Find the alternative with the highest average outcome
- Pick the outcome with the maximum number
- Assumes each state of nature is equally likely to occur

Decision Making Under Risk

- Each possible state of nature has an assumed probability
- States of nature are mutually exclusive
- Probabilities must sum to 1
- Determine the **expected monetary value (EMV)** for each alternative



Do you remember?



Activity

20 minutes

- Do you remember “the best” and “the worst” decisions you have ever made?
- Give yourself a few minutes to remember
- And share your answers with your peers



Expected Monetary Value

$$\begin{aligned} \text{EMV (Alternative } i) = & (\text{Payoff of 1}^{\text{st}} \text{ state of nature}) \\ & \times (\text{Probability of 1}^{\text{st}} \text{ state of nature}) \\ & + (\text{Payoff of 2}^{\text{nd}} \text{ state of nature}) \\ & \times (\text{Probability of 2}^{\text{nd}} \text{ state of nature}) \\ & + \dots + (\text{Payoff of last state of nature}) \times (\text{Probability of last state of nature}) \end{aligned}$$

Decision Making Under Certainty

- Is the cost of perfect information worth it?
- Determine the expected value of perfect information (EVPI)

Expected Value of Perfect Information

EVPI is the difference between the payoff under certainty and the payoff under risk

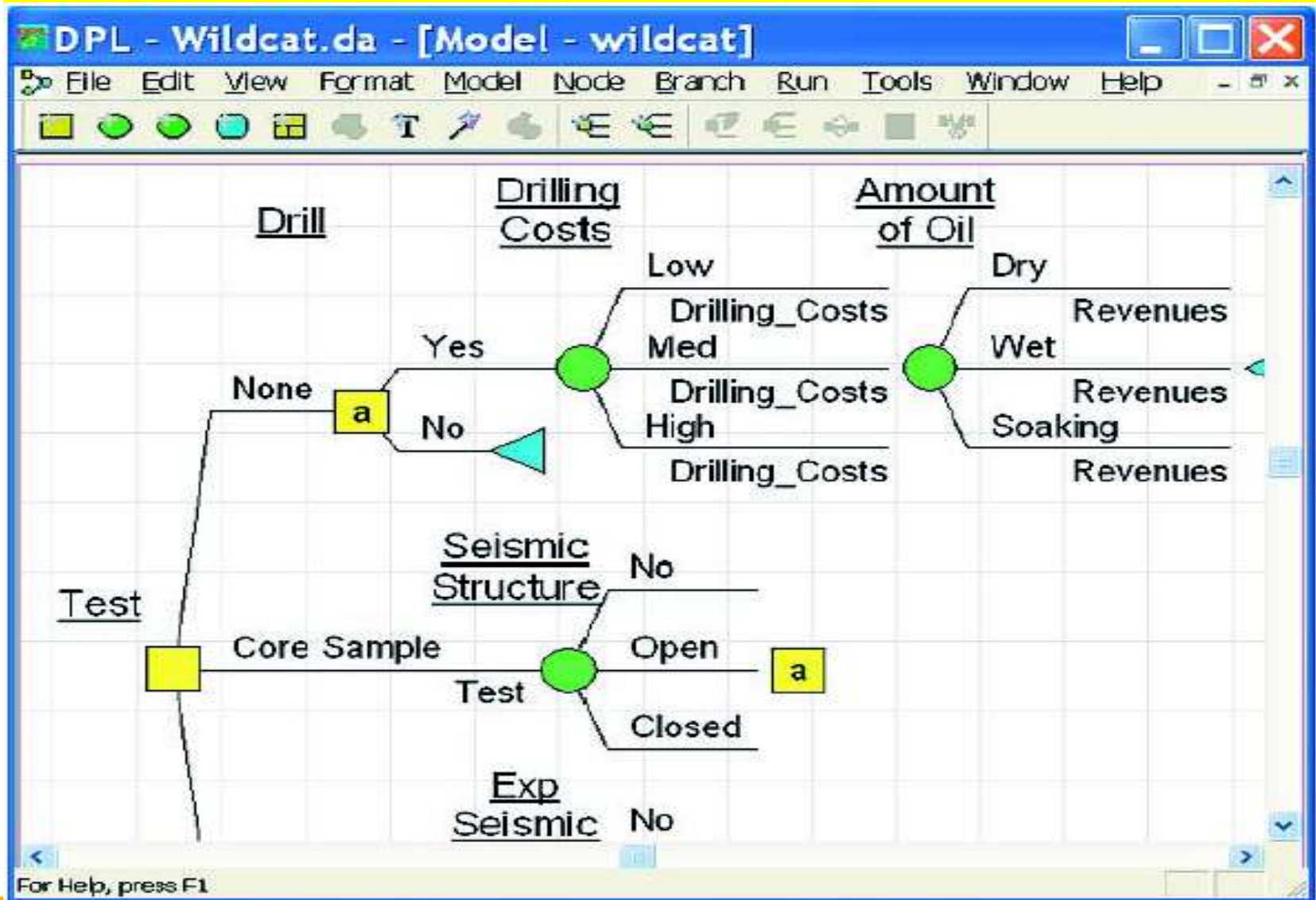
$$\text{EVPI} = \begin{array}{c} \text{Expected value} \\ \text{with perfect} \\ \text{information} \end{array} - \begin{array}{c} \text{Maximum} \\ \text{EMV} \end{array}$$

Expected value *with* perfect information (EVwPI) = (Best outcome or consequence for 1st state of nature) x (Probability of 1st state of nature)
+ Best outcome for 2nd state of nature) x (Probability of 2nd state of nature)
+ ... + Best outcome for last state of nature) x (Probability of last state of nature)

Decision Trees

- Information in decision tables can be displayed as decision trees
- A decision tree is a graphic display of the decision process that indicates decision alternatives, states of nature and their respective probabilities, and payoffs for each combination of decision alternative and state of nature
- Appropriate for showing sequential decisions

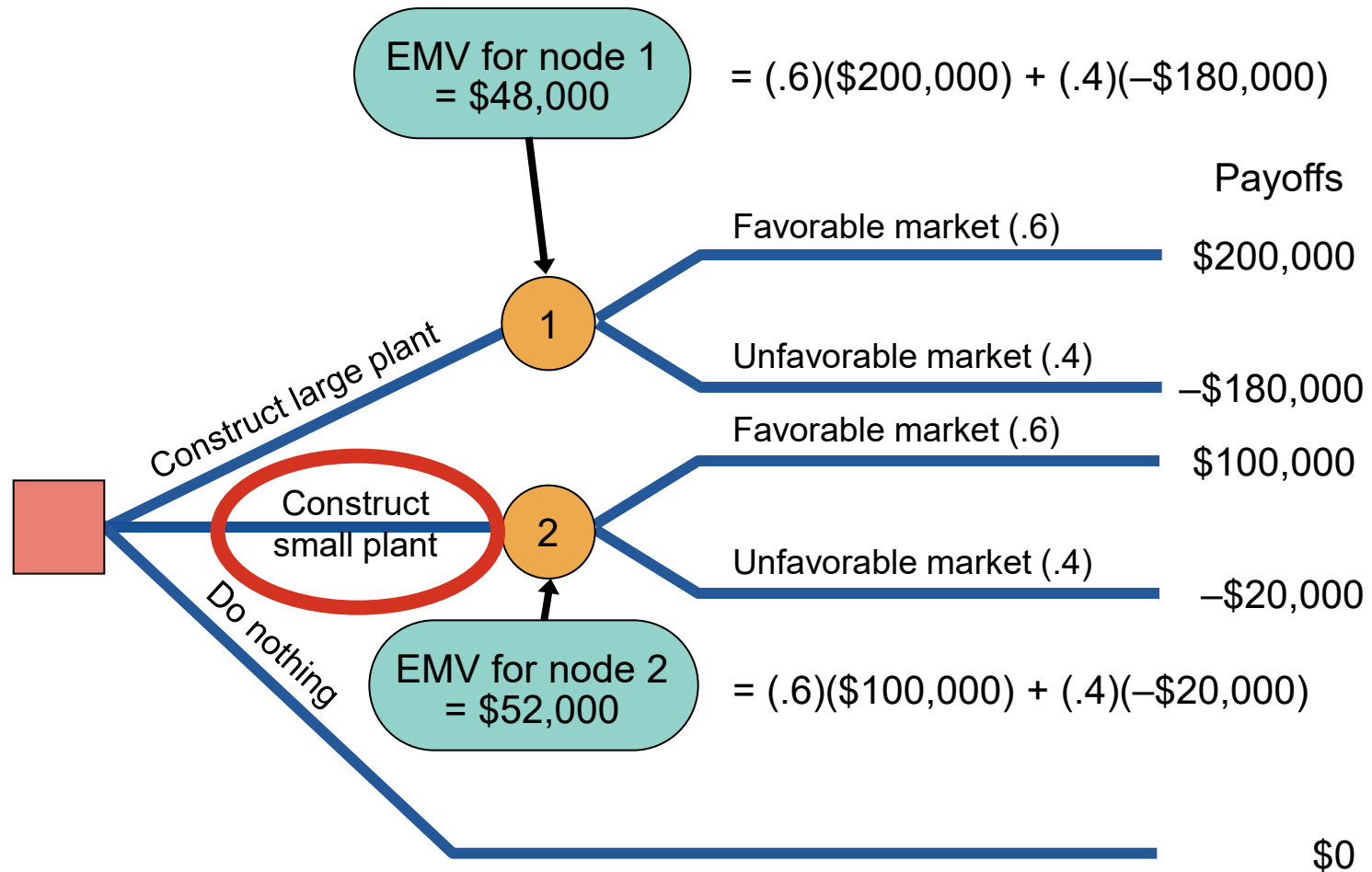
Decision Trees



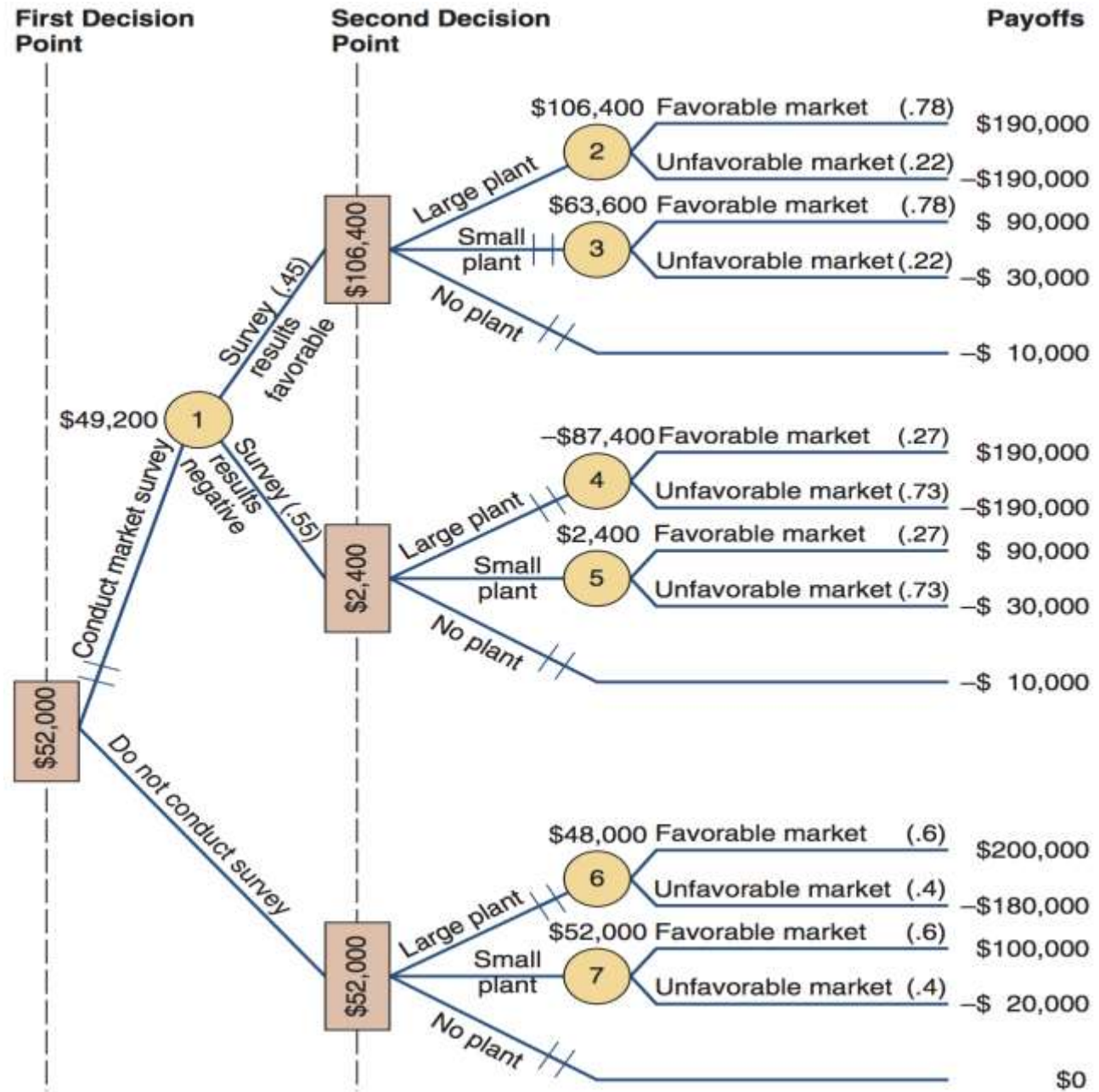
Decision Trees

1. Define the problem
2. Structure or draw the decision tree
3. Assign probabilities to the states of nature
4. Estimate payoffs for each possible combination of decision alternatives and states of nature
5. Solve the problem by working backward through the tree computing the EMV for each state-of-nature node

Decision Tree Example



Complex Decision Tree Example



Draw Your Own Decision Tree Activity

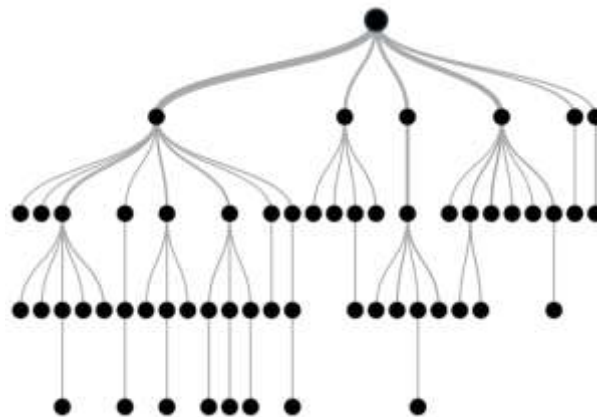


Activity

20 minutes

Decision trees for your best and worst decisions

Now draw your decision trees for the best/worst decisions discussed earlier and use it to illustrate your answers to the class



Independent study



Read the article on ‘Decision making in product design’

Assignment support



Traffic Light *it*

Where is your learning at?



Activity

5 minutes



Red = I have misunderstood some of the topics. (write down 2 questions that you need help with)

Amber = I have understood most of the topics so far (write down 1 question and 1 key theme you have understood)

Green = I have understood the topics so far (write down 2 key themes from the lesson to show your understanding)



Time for Reflection...



Activity

10 minutes

Reflect on what you have learned today and
share on AULA/Community Space:

What happened?

What were you thinking?

What were you feeling?

What was good / not good about today's
lecture?

Will it be useful?

A] Take down at least two points that you learned new

B] How you are going to transfer this learning in your CW
and/or workplace



*Thank
You*

Essential Reading

- Slack, N., Brandon-Jones, A. and Johnston, R. (2016) Operations Management. 8th edn. Harlow: Pearson
- Torrington, D., Hall, L., Atkinson, C. and Taylor, S. (2017) Human Resource Management. 10th edn. Harlow: Pearson

Recommended Reading

- Slack, N., Brandon-Jones, A., Johnston, R. and Betts, A. (2015) Operations and Process Management: Principles and Practice for Strategic Impact. 4th edn. Harlow: Pearson
- Bratton, J. and Gold, J. (2012) Human Resource Management: Theory and Practice. 5th edn. Basingstoke: Palgrave MacMillan
- Heizer, J., Render, B. and Munson, C. (2017) Principles of Operations Management: Sustainability and Supply Chain Management. 10th edn. Harlow: Pearson
- CMI Website : Management Direct

Session Reference List

- Bozarth, C. and Handfield, R. (2013). Introduction to Operations and Supply Chain Management. 3rd ed. Essex: Pearson Education, pp.24, 25, 28, 270, 271, 475.
- Greasley, A. (2013) Operations Management. 3rd edn. John Wiley & Sons
- Heizer, J., Render, Barry, author, & Munson, Chuck, author. (2017). Principles of operations management sustainability and supply chain management (Tenth edition..; Global ed.).
- Hill, A. and Hill, T. (2018) Essential operations management. Second edition. London: Palgrave
- Slack, N., Brandon-Jones, Alistair, author, & Johnston, Robert, author. (2013). Operations management (Seventh ed.). Harlow, England: Pearson
- Slack, N. et al. (2015) Operations and process management: principles and practice for strategic impact. Fourth edition. Harlow, England: Pearson
- Slack, N. and Brandon-Jones, A. (2018) Operations and process management: principles and practice for strategic impact. Fifth edition. Harlow, England: Pearson